

Accelerating physics-informed neural networks for full waveform inversion using a hybrid quantum-classical finite-basis architecture

Hoang Anh Nguyen^{a,*}, Divakar Vashisth^b, and Ali Tura^a

^aDepartment of Geophysics, Colorado School of Mines, Golden, CO 80401, USA

^bDepartment of Energy Science and Engineering, Stanford University, Stanford, CA 94305, USA

*Corresponding author: hoanganh.nguyen@mines.edu

May 25, 2026

Abstract

Full waveform inversion (FWI) reconstructs heterogeneous material properties from receiver data but remains computationally demanding. Physics-informed neural networks (PINNs) and their domain-decomposed variants (FBPINNs) offer a mesh-free alternative but face convergence challenges when representing complex velocity fields. We present a hybrid quantum-classical FBPINN for acoustic FWI, in which the velocity network is implemented as a classical-to-quantum pipeline terminating in a parameterized quantum circuit (PQC). The PQC is realized as a differentiable JAX statevector simulator, enabling end-to-end automatic differentiation through the classical PINN, the quantum circuit, and the physics-informed loss. On a geophysical anomaly benchmark, the quantum hybrid reaches a lower L1 velocity error than the primary classical FBPINN baseline in approximately $8\times$ fewer training iterations, despite using $\sim 33\%$ fewer trainable parameters, and it outperforms all 15 classical hyperparameter variants tested. A second benchmark (checkerboard) demonstrates the generality of the inversion pipeline, confirming that the quantum hybrid architecture successfully resolves higher-frequency structural variations. Our framework is broadly applicable to wave-based inverse problems beyond geophysics, including medical ultrasound tomography and non-destructive evaluation.

Keywords: Full Waveform Inversion, Wave Propagation, Quantum Computing, Parameterized Quantum Circuit, PINNs

1 Introduction

Full waveform inversion (FWI) is a powerful technique for reconstructing material properties from recorded waveform data by exploiting the full information content of wave signals, including amplitude, phase, and frequency. Originally developed in the context of exploration geophysics for subsurface velocity imaging¹, FWI has since become central to a wide range of disciplines. In global seismology, it enables high-resolution imaging of crustal and mantle structures^{2,3}; in hydrocarbon exploration and CO₂ monitoring, it provides detailed velocity models for reservoir characterization⁴ and is increasingly applied to high-density datasets acquired via distributed acoustic sensing⁵. Beyond geophysics, FWI principles have found growing applications in medical imaging, where ultrasound computed tomography employs waveform inversion to reconstruct tissue properties for breast cancer detection and characterization^{6,7}, and in photoacoustic tomography, where wave-based inversion recovers optical absorption maps from acoustic measurements⁸. Non-destructive evaluation of engineering structures similarly leverages full waveform methods to detect defects and characterize material properties⁹. Despite its versatility across these domains, FWI remains computationally demanding, requiring iterative solutions of the

forward wave equation and the computation of gradients through adjoint-state methods, often at substantial cost for large-scale and high-frequency problems.

Physics-informed neural networks (PINNs) have emerged as a mesh-free alternative for solving partial differential equations (PDEs) by minimizing PDE residuals at collocation points¹⁰, approximating solutions without traditional numerical discretization. This paradigm has been extended to inverse problems, where unknown model parameters are optimized alongside the solution. Recent works have applied PINNs to wave-related problems^{11–13}, offering a fundamentally different approach to the classical adjoint-state framework. In particular, Rasht-Behesht et al.¹⁴ demonstrated the application of PINNs to FWI, achieving accurate velocity reconstruction from receiver data by jointly optimizing the wavefield and velocity networks. However, standard PINNs face well-documented challenges when applied to problems involving high-frequency wave propagation, multi-scale physics, and large computational domains. The spectral bias of neural networks, their tendency to learn low-frequency features before high-frequency ones, can severely limit convergence in wave propagation problems. To address these limitations, finite-basis physics-informed neural networks (FBPINNs) were introduced by Moseley et al.¹⁵, employing a domain decomposition strategy in which the global solution is represented as a sum of locally supported basis functions, each parameterized by an independent neural network. This approach effectively mitigates spectral bias, improves parallel scalability, and enables the solution of problems that are intractable for standard PINNs. To date, however, no work has applied the domain decomposition advantages of FBPINNs to waveform inversion.

In parallel with advances in scientific machine learning, quantum computing has attracted growing interest for its potential to accelerate computational tasks across diverse domains. Parameterized quantum circuits (PQCs), also known as variational quantum circuits (VQCs), have emerged as a near-term approach to quantum machine learning, operating within the constraints of current noisy intermediate-scale quantum (NISQ) devices¹⁶. PQCs encode classical data into quantum states through feature maps, process the information through parameterized unitary operations, and extract classical outputs via measurement. Several studies have explored the integration of quantum circuits with neural networks, forming hybrid quantum-classical architectures that leverage the expressive power of quantum Hilbert spaces while maintaining the trainability and flexibility of classical networks^{17–19}.

The intersection of quantum computing and PINNs has begun to receive attention, with recent works investigating quantum physics-informed neural networks (QPINNs) for solving differential equations²⁰. Leong et al.²¹ benchmarked hybrid quantum physics-informed neural networks (HQPINNs) for high-speed flow problems, demonstrating that these architectures offer competitive accuracy and stability with a reduced parameter cost compared to classical PINNs. While they found that pure quantum models struggle with non-harmonic features like shocks, the hybrid approach successfully mitigates these artifacts by leveraging classical sub-networks. Similarly, Berger et al.²² proposed trainable embedding quantum physics-informed neural networks (TE-QPINNs) for solving nonlinear PDEs, including the Navier-Stokes equations. Their approach achieved superior results compared to classical PINNs while maintaining the same number of parameters, highlighting the potential for more efficient optimization in high-dimensional parameter spaces. These studies suggest that quantum circuits can serve as powerful function approximators within the PINN framework, potentially offering advantages in expressivity and parameter efficiency for forward PDE problems. When extending quantum methods to geophysical inverse problems, research has thus far primarily relied on alternative computing paradigms. For instance, quantum annealing has recently been applied to seismic traveltime tomography^{*23}, where the inversion is cast as a linear least-squares problem, mapped to a quadratic unconstrained binary optimization (QUBO) problem, and solved on a quantum annealer. However, the application of gate-based hybrid quantum-classical neural architectures to highly nonlinear inverse problems like FWI, where both the wavefield and the velocity model are represented by learned networks and coupled through the acoustic wave equation, remains

largely unexplored.

In this work, we present a hybrid quantum-classical FBPINN for acoustic FWI. The framework retains classical fully-connected networks for the wavefield representation across decomposed subdomains and introduces a PQC into the velocity network via a classical-to-quantum pipeline: classical embedding layers map spatial coordinates into a low-dimensional latent space matched to the number of qubits, the PQC processes the encoded features through parameterized rotations and entangling gates, and a classical output layer rescales the bounded circuit output into physical velocity units. To isolate the architectural benefits of the quantum integration from the hardware noise of current near-term devices, the PQC is implemented as a JAX-native statevector simulator. This approach enables exact, end-to-end automatic differentiation through the classical PINN, the quantum circuit, and the physics-informed loss, avoiding the prohibitive computational overhead of parameter-shift rules required on physical hardware. This design combines the scalability advantages of FBPINN domain decomposition with the structured, bounded expressivity of a PQC-based velocity parameterization. We demonstrate the efficacy of this architecture on a geophysical anomaly benchmark. The hybrid model achieves $\sim 8\times$ faster convergence in training iterations and a lower final L1 velocity error than the primary classical FBPINN baseline, despite using $\sim 33\%$ fewer trainable parameters, and it consistently outperforms all 15 alternative classical hyperparameter variants tested. In the following sections, we present the comparative performance of our hybrid architecture, discuss its broader implications for scientific machine learning, and detail the underlying mathematical and algorithmic methodology.

2 Results

To systematically assess the advantages of the proposed hybrid architecture, we evaluate its inversion performance on a canonical wave-propagation test case: reconstructing a spatial velocity anomaly from surface receiver recordings. By comparing this hybrid approach against a purely classical baseline under identical training conditions, we aim to isolate the specific impact of the parameterized quantum circuit on convergence speed, model expressivity, and parameter efficiency.

2.1 Architecture and training setup

To establish a rigorous comparison, we evaluate two distinct configurations applied to the identical FWI problem: a purely classical FBPINN baseline and our proposed hybrid quantum-classical architecture. In the classical baseline, the velocity field $v(\mathbf{x})$ is parameterized by a standard fully-connected neural network. Conversely, the hybrid model replaces this component with a classical-to-quantum pipeline that terminates in a four-qubit parameterized quantum circuit (PQC).

Beyond this velocity network substitution, both configurations share a unified foundation. They employ the same wavefield network $\phi(\mathbf{x}, t)$, identical FBPINN subdomain decomposition strategies ($5 \times 3 \times 5$ grid with a 0.35 overlap), and an equivalent physics-informed composite loss function incorporating the PDE residual, wavefield snapshot initial conditions (ICs), data misfit, and free-stress boundary constraints. Notably, the quantum hybrid model achieves its expressivity using significantly fewer trainable parameters. Schematic representations of the hybrid framework (Figure 1) and the PQC topology (Figure 2) are provided alongside a summary of the network architectures in Table 1. Methodological details, including the underlying acoustic wave equations and loss derivations, are deferred to the Methods section.

2.2 Anomaly benchmark

Our primary evaluation utilizes the velocity model established by Rasht-Behesht et al.¹⁴, which defines a 2D acoustic domain ($1.15 \text{ km} \times 0.45 \text{ km}$) featuring a localized low-velocity anomaly

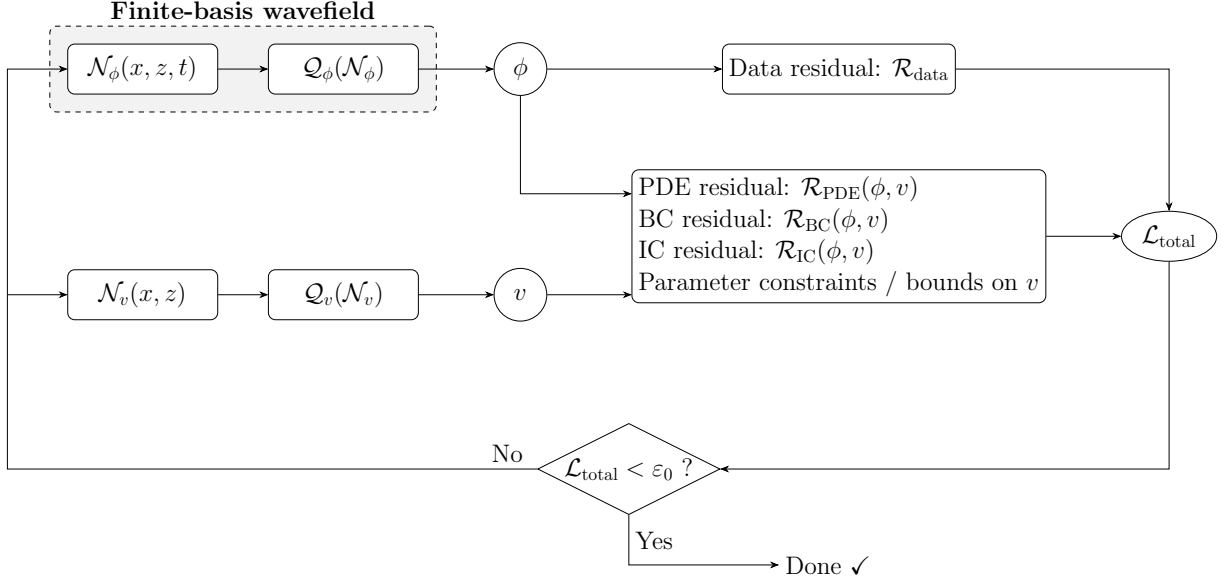


Figure 1: Schematic of the finite-basis classical-quantum hybrid architecture, combining classical neural networks and quantum circuits with physics-informed constraints. $\mathcal{N}$ denotes classical networks, $\mathcal{Q}$ quantum circuits, ϕ the predicted wavefield, v the inverted velocity, and $\mathcal{R}$ the physics- and data-based residuals.

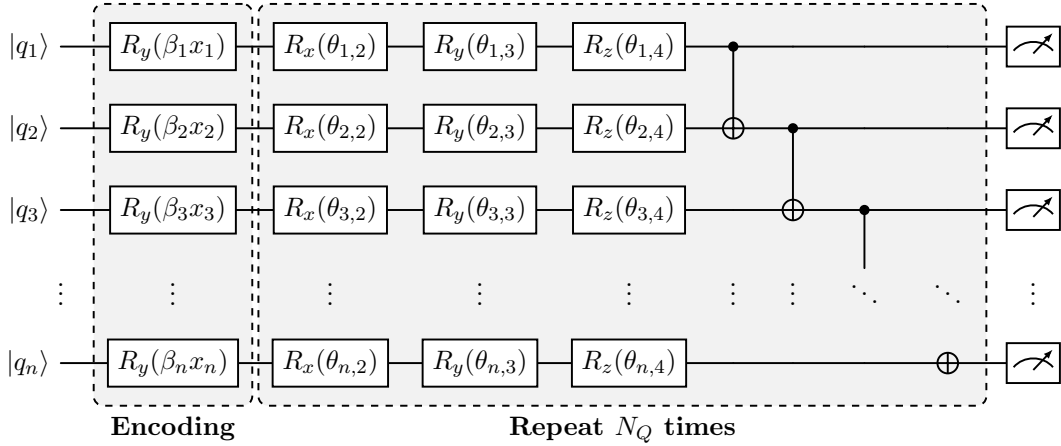


Figure 2: Multilayer PQC $\mathcal{Q}$. Each classical input feature x_i is angle-encoded via an $R_y(\beta_i x_i)$ rotation (encoding layer, grey), where β_i is a trainable scaling parameter. Variational layers of single-qubit rotations R_x , R_y , R_z followed by a linear CNOT ladder are repeated N_Q times (grey block). Each qubit is measured in the Pauli-Z basis, and the outputs are averaged to yield a scalar in $[-1, 1]$.

(~ 2.5 km/s embedded within a ~ 3.0 km/s background; Figure 3a). Ground-truth receiver data are generated through a spectral element method^{*24} forward simulation driven by a Gaussian source at (0.35, 0.19) km. The inversion process is constrained by 500 ms of wavefield displacements recorded by a vertical array of 20 receivers positioned along the right boundary ($x = 1.15$ km), supplemented by two early displacement snapshots ($t_0 = 0.100$ s and $t_1 = 0.115$ s). This transmission-dominated configuration, featuring a single source and a one-sided receiver array, represents a severely ill-posed inverse problem. The limited angular illumination inherently restricts the recovery of high-wavenumber spatial components, rigorously testing the regularizing capacity of the chosen neural architecture.

Table 1: Network configurations for the anomaly benchmark. $\mathcal{Q}(n_q^{\times n_L})$ denotes a PQC with n_q qubits and n_L layers. Subdomain decomposition ($5 \times 3 \times 5$, overlap 0.35) applies to the wavefield network ϕ ; the velocity network v is a single global network. Loss weights: $\lambda_{\text{PDE}}=0.1$, $\lambda_{\text{IC}}=1.0$, $\lambda_{\text{seis}}=1.0$, $\lambda_{\text{BC}}=0.1$. Learning rate 10^{-4} . See Supplementary Table S1 for the full hyperparameter variant sweep.

Model	ϕ network $(x, z, t) \rightarrow \phi$	c network $(x, z) \rightarrow v$	# Params
classical	$\mathcal{N}: 32^{\times 2} \rightarrow 16^{\times 2}$	$\mathcal{N}: 20^{\times 5}$	151,836
quantum	$\mathcal{N}: 32^{\times 2} \rightarrow \mathcal{Q}: 4^{\times 2}$	$\mathcal{N}: 20^{\times 3} \rightarrow \mathcal{Q}: 4^{\times 2}$	101,812

Starting from a purely homogeneous initial guess (Figure 3b), both the classical and hybrid architectures successfully localize the low-velocity anomaly without suffering from the cycle-skipping artifacts that frequently plague classical adjoint-state FWI under poor starting models. As illustrated by the inverted velocity field of the quantum hybrid model (Figure 3c), the recovered anomaly closely tracks the true model’s spatial distribution and peak amplitude. The minor lateral smearing observed in the reconstruction strictly aligns with the theoretical limits of resolving power imposed by the single-source geometry, confirming that the network has extracted the maximum available information from the receiver data.

The most profound distinction between the classical and quantum approaches emerges in their optimization dynamics. Figure 3d tracks the L1 velocity error against training iterations, revealing structurally different convergence trajectories. The quantum hybrid model undergoes a rapid initial descent, achieving an L1 error of 2.4×10^{-2} in merely 65,000 iterations. In stark contrast, the classical baseline exhibits prolonged plateauing behavior, requiring 500,000 iterations to reach a comparable, albeit slightly higher, final error of 2.9×10^{-2} . This constitutes an approximate $8\times$ acceleration in the necessary optimization steps, achieved despite the hybrid model operating with a significantly lower number of trainable parameters (101,812 vs. 151,836). Because the PQC is simulated classically on GPU hardware without leveraging true quantum superposition or entanglement for computational speedup, we attribute this acceleration exclusively to the highly favorable loss landscape induced by the quantum-circuit parameterization.

To rigorously confirm that this performance gap is fundamentally architectural rather than an artifact of sub-optimal classical tuning, we conducted an exhaustive ablation study comprising 15 distinct classical FBPINN variants. These baselines spanned a wide spectrum of configurations, including severely over-parameterized velocity networks (up to 60×3 hidden layers), alternative subdomain partitions ($3 \times 2 \times 3$ and $8 \times 4 \times 8$), varying learning rates, and adjusted physical loss weightings (detailed comprehensively in Supplementary Table S1). As depicted by the thin colored traces in Figure 3d, simply increasing classical capacity or tuning hyperparameters failed to bridge the gap. None of the 15 classical configurations could match the steep descent or the final L1 error of the quantum hybrid trajectory at any stage during the 500,000 iterations. This confirms that the convergence advantage is a robust consequence of mapping the spatial coordinates through the PQC’s structured Hilbert space.

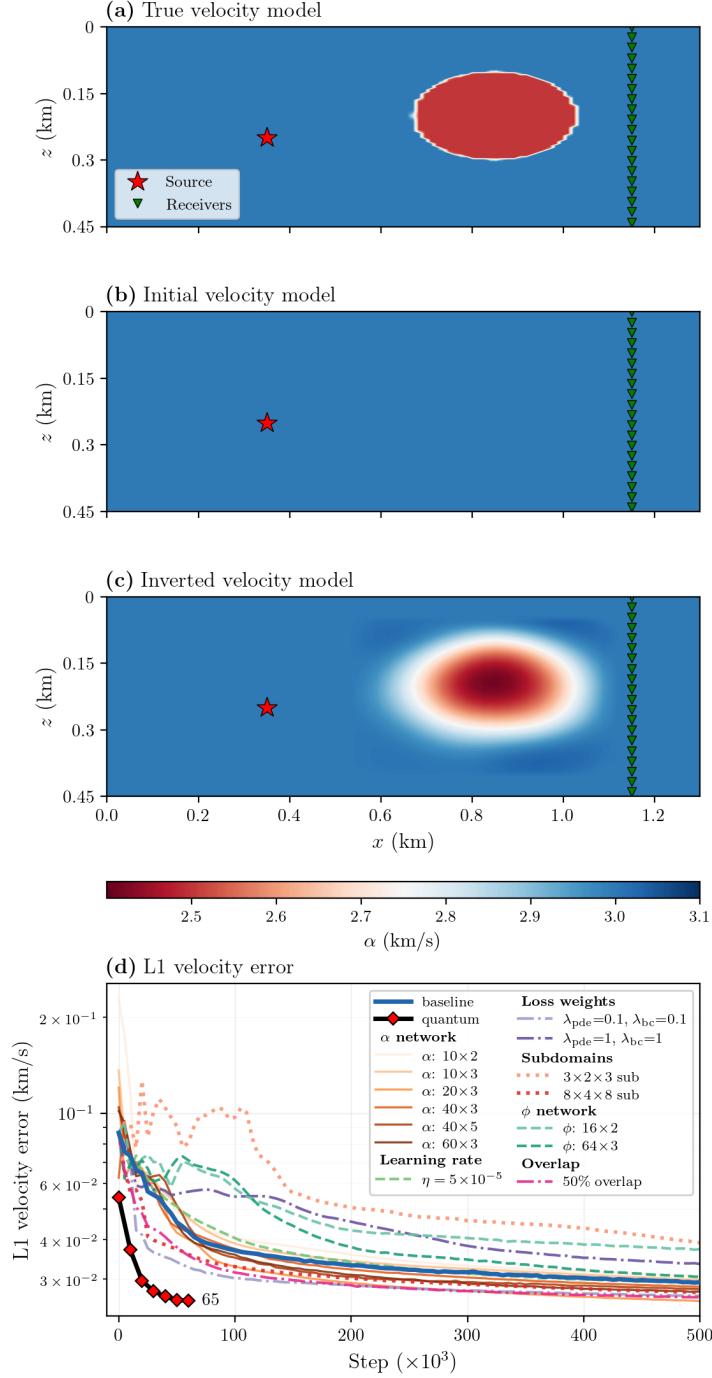


Figure 3: Anomaly benchmark inversion and convergence. (a) True velocity model used for the SPECFEM2D forward simulation, with source (red star) and receiver array (green triangles). (b) Initial homogeneous velocity model. (c) Inverted velocity model from the quantum hybrid FBPINN at 65,000 training iterations. (d) L1 velocity error versus training iteration for the classical FBPINN baseline (blue), the quantum hybrid (red markers labelled 65k), and 15 classical hyperparameter variants (thin coloured lines). The quantum hybrid reaches a lower L1 error than any classical variant in approximately $8\times$ fewer training iterations.

2.3 Checkerboard benchmark

To further probe the resolution limits of the inversion framework, we test a checkerboard velocity model. This configuration introduces a complex 3×3 grid of alternating ± 0.4 km/s perturbations

against a 3.0 km/s background, while maintaining the identical domain and source-receiver geometry used in the anomaly benchmark (Figure 4a).

The classical FBPINN successfully resolves the checkerboard pattern after 2×10^6 training iterations (Figure 4b). As expected from the limited angular coverage of the benchmark’s geometry, the structural resolution gradually degrades for blocks situated further from the receiver array. The quantum hybrid architecture similarly resolves the checkerboard pattern. It employs the identical network configurations established for the anomaly benchmark, with the training regimen adapted to include a higher density of collocation points necessary to adequately constrain the smaller-scale checkerboard structures.

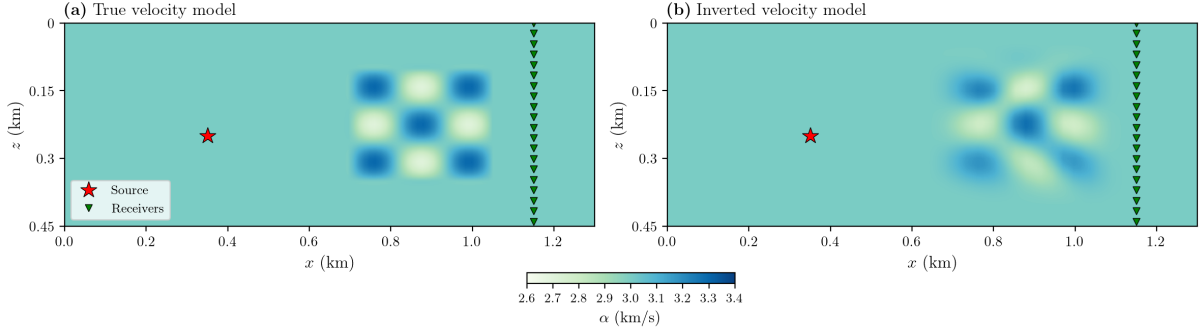


Figure 4: Checkerboard benchmark inversion. (a) True velocity model: a 3×3 checkerboard of alternating ± 0.4 km/s perturbations on a 3.0 km/s background, with source (red star) and receiver array (green triangles). (b) Inverted velocity model from the classical FBPINN at 2×10^6 training iterations.

3 Discussion

The quantum hybrid velocity network reaches a lower L1 velocity error than the primary classical FBPINN baseline in approximately $8 \times$ fewer training iterations, despite using $\sim 33\%$ fewer trainable parameters. Furthermore, it outperforms all 15 alternative classical hyperparameter variants tested across a wide range of parameter counts. We attribute this to three properties of the classical-to-PQC pipeline ^{*25}. First, despite its smaller parameter count, the PQC operates in an exponentially large Hilbert space ^{*26,27}: an n -qubit statevector has 2^n complex amplitudes (for $n = 4$, a 16-dimensional complex space, i.e. 32 real degrees of freedom per evaluation), and the measured expectation value

$$f(\mathbf{x}; \boldsymbol{\beta}, \boldsymbol{\theta}) = \frac{1}{n} \sum_{i=0}^{n-1} \langle \psi(\mathbf{x}; \boldsymbol{\beta}, \boldsymbol{\theta}) | Z_i | \psi(\mathbf{x}; \boldsymbol{\beta}, \boldsymbol{\theta}) \rangle \quad (1)$$

is a nonlinear function of all variational angles acting jointly on this high-dimensional representation. A classical MLP with a comparable parameter count operates in a much smaller latent space and must spend its parameters on both feature construction and output generation. Second, angle encoding endows the PQC output with a specific finite-frequency Fourier structure ^{*28},

$$f(\mathbf{x}) = \sum_{\boldsymbol{\omega} \in \Omega} c_{\boldsymbol{\omega}}(\boldsymbol{\theta}) e^{i\boldsymbol{\omega} \cdot (\boldsymbol{\beta} \odot \mathbf{x})}, \quad (2)$$

in which the trainable angles control the Fourier coefficients $c_{\boldsymbol{\omega}}$ while the accessible frequency set Ω is fixed by the eigenvalues of the encoding generators. This built-in spectral prior may match smooth, spatially-correlated velocity fields more efficiently than an unconstrained tanh

MLP, which has no such structure by default. Third, the PQC output is bounded to $[-1, 1]$ by construction, which hard-codes a range constraint on the velocity perturbation (equation 14) and removes an avenue for early training to wander into pathological high-amplitude solutions; entangling CNOT gates additionally introduce non-separable correlations between the spatial inputs that a product-structured MLP only recovers through repeated tensor contractions. The robustness of the advantage across classical hyperparameter choices (Figure 3d) argues against an unlucky baseline, though we emphasize that these gains are in optimization iterations on classically-simulated circuits and do not constitute a quantum computational advantage.

Our work complements recent hybrid quantum-PINN studies in other PDE-constrained settings. Leong et al.²¹ benchmarked hybrid quantum PINNs for high-speed compressible flow, finding competitive accuracy at reduced parameter cost compared with classical PINNs. Berger et al.²² introduced trainable-embedding quantum PINNs for nonlinear PDEs including the Navier-Stokes equations, again observing efficiency gains at matched parameter count. The present work differs in two ways. First, we target a coupled forward-inverse problem (waveform inversion) rather than a pure forward PDE; the velocity field is learned jointly with the wavefield. Second, we integrate the PQC within a domain-decomposed FBPINN¹⁵, extending the PINN-based FWI formulation of Rasht-Behesht et al.¹⁴ to handle larger and higher-frequency inversion problems via overlapping subdomain networks.

Several limitations should be noted. (i) The present experiments are two-dimensional and acoustic. Extending to three-dimensional elastic inversion is the natural next step but is constrained by the $\mathcal{O}(2^n)$ memory cost of statevector simulation as qubit counts grow. Scaling to larger circuits on real hardware further raises the concern of barren plateaus, vanishingly small gradients in high-depth parameterized circuits^{*29}, which would require careful initialization and ansatz choice. (ii) Inversions use a single source; production FWI pipelines incorporate tens to thousands of sources with different illuminations, and PINN-based FWI in particular is known to be sensitive to loss-landscape pathologies under data sparsity^{*30}. (iii) The quantum circuit is simulated classically. On real NISQ hardware, gradients would be computed via the parameter-shift rule (two circuit evaluations per parameter), and device noise could erode the iteration-level advantage observed here.

Looking forward, the small qubit counts used here ($n = 4$, $L = 2$) are compatible with near-term quantum hardware, which suggests that genuine hardware demonstrations are feasible once noise-mitigation techniques reach sufficient maturity. Systematic studies at matched parameter counts across circuit depths and widths would clarify how the convergence advantage scales with quantum-circuit capacity. Beyond geophysics, the framework is directly applicable to any wave-based inverse problem expressible as a physics-informed learning task, including medical ultrasound tomography^{6,7}, photoacoustic imaging⁸, and non-destructive evaluation⁹, and we expect the same convergence benefits to transfer to those settings.

4 Methods

4.1 Acoustic wave equation

We consider two-dimensional acoustic wave propagation in a heterogeneous medium with spatially varying wave speed $v(\mathbf{x})$ and constant density. With no explicit source injection, the governing equation is the source-free acoustic wave equation,

$$\frac{\partial^2 u(\mathbf{x}, t)}{\partial t^2} = v^2(\mathbf{x}) \nabla^2 u(\mathbf{x}, t), \quad (3)$$

where $u(\mathbf{x}, t)$ is a scalar potential whose spatial derivatives ($\partial u / \partial x$, $\partial u / \partial z$) give the two components of the displacement field. Wave propagation is initiated by prescribing the wavefield at two closely-spaced instants, $t_0 = 0.100$ s and $t_1 = 0.115$ s, using snapshots of the displacement field taken from a spectral-element simulation; both snapshots are imposed as soft constraints in the

loss function. A free-stress boundary condition (vanishing pressure Laplacian) is enforced on the top surface of the physical domain. This formulation provides a well-posed initial-boundary-value problem without requiring an explicit source time function.

4.2 Finite-basis physics-informed neural network

PINNs¹⁰ approximate solutions of PDEs by a neural network $u_\theta(\mathbf{x})$ and train θ to minimize the PDE residual at a set of collocation points $\{\mathbf{x}_i\}_{i=1}^N$ drawn from the domain. Writing the governing PDE in residual form $\mathcal{N}[u](\mathbf{x}) = 0$, the PDE loss is

$$\mathcal{L}_{\text{PDE}} = \frac{1}{N} \sum_{i=1}^N |\mathcal{N}[u_\theta(\mathbf{x}_i)]|^2, \quad (4)$$

with spatial and temporal derivatives computed by automatic differentiation of the network with respect to its inputs. Boundary conditions, initial conditions, and observational data are added as additional weighted loss terms (the full composite loss used for FWI is given in the Hybrid architecture subsection below). Standard PINNs, however, are known to suffer from spectral bias, a tendency to learn low-frequency features before high-frequency ones, which severely limits their effectiveness in multi-scale wave problems such as FWI, where both the velocity field and the wavefield exhibit sharp transitions and oscillatory structure.

To overcome this limitation, we adopt the FBPINN framework of Moseley et al.¹⁵. The global domain Ω is partitioned into K overlapping rectangular subdomains $\{\Omega_k\}$, each equipped with an independent neural network $u_{\theta_k}(\mathbf{x})$. The global solution is reconstructed as a partition-of-unity sum,

$$u(\mathbf{x}) = \sum_{k=1}^K w_k(\mathbf{x}) u_{\theta_k}(\mathbf{x}), \quad (5)$$

where the window functions w_k are smooth cosine bumps supported on Ω_k with $\sum_k w_k(\mathbf{x}) = 1$ for all $\mathbf{x} \in \Omega$. Each local network only needs to represent the solution's behaviour on its own subdomain, with two consequences: (i) the local frequency content each network must learn is bounded by the subdomain size, alleviating spectral bias; and (ii) training is embarrassingly parallel across subdomains, enabling batched evaluation via JAX's `vmap`. The physics-informed loss is evaluated on the reconstructed global solution, so gradients flow through the window-weighted sum back to the individual subdomain networks.

For the wavefield $\phi(x, z, t)$ we use a three-dimensional subdomain decomposition (n_x, n_z, n_t) , $5 \times 3 \times 5$ in the anomaly baseline with subdomain overlap 0.35. The velocity network $v(x, z)$ is not decomposed, it is a single global network, since the velocity field typically varies more smoothly than the wavefield and does not require local representation.

4.3 Parameterized quantum circuits

This section introduces the quantum computing concepts required to understand the hybrid classical-quantum architecture presented in the Hybrid architecture subsection of Methods.

4.3.1 Qubits and quantum states

The fundamental unit of quantum information is the qubit, a two-level quantum system whose state is represented by a unit vector in $\mathbb{C}^2$. In the computational basis $\{|0\rangle, |1\rangle\}$, an arbitrary single-qubit state is written as

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle, \quad |\alpha|^2 + |\beta|^2 = 1, \quad (6)$$

where $\alpha, \beta \in \mathbb{C}$ are probability amplitudes. Geometrically, the state of a single qubit can be visualised as a point on the Bloch sphere, parameterised by polar and azimuthal angles. An n -qubit register is described by a statevector in $\mathbb{C}^{2^n}$,

$$|\psi\rangle = \sum_{k=0}^{2^n-1} \alpha_k |k\rangle, \quad \sum_{k=0}^{2^n-1} |\alpha_k|^2 = 1, \quad (7)$$

where $|k\rangle$ denotes the k -th computational basis state. The exponential growth of the Hilbert space dimension with qubit count underlies both the potential computational advantage of quantum systems and the cost of their classical simulation.

4.3.2 Quantum gates

Quantum computation proceeds by applying quantum gates, unitary transformations that evolve the statevector while preserving its norm. The circuit used in this work employs two kinds of gate.

The single-qubit rotations $R_x(\theta)$, $R_y(\theta)$, $R_z(\theta)$ are parameterized unitaries

$$R_k(\theta) = e^{-i\theta \sigma_k/2}, \quad k \in \{x, y, z\}, \quad (8)$$

where σ_k are the Pauli matrices. Written explicitly,

$$R_x(\theta) = \begin{pmatrix} \cos \frac{\theta}{2} & -i \sin \frac{\theta}{2} \\ -i \sin \frac{\theta}{2} & \cos \frac{\theta}{2} \end{pmatrix}, \quad R_y(\theta) = \begin{pmatrix} \cos \frac{\theta}{2} & -\sin \frac{\theta}{2} \\ \sin \frac{\theta}{2} & \cos \frac{\theta}{2} \end{pmatrix}, \quad R_z(\theta) = \begin{pmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{pmatrix}. \quad (9)$$

Each gate rotates the single-qubit state by angle θ about the corresponding axis of the Bloch sphere, and the composition $R_z(\theta_3) R_y(\theta_2) R_x(\theta_1)$ is a general single-qubit unitary with three independent parameters, capable of mapping the initial state to any point on the Bloch sphere.

The entangling gate is the controlled-NOT (CNOT), a two-qubit unitary that flips the target qubit if and only if the control qubit is in state $|1\rangle$:

$$\text{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}. \quad (10)$$

CNOT is the sole multi-qubit gate used in our circuit and the sole source of entanglement, correlations between qubits that have no classical analogue, which enables the PQC to represent functions of multiple input features that cannot be factorized across individual qubits.

4.3.3 PQC ansatz

A PQC³¹, also referred to as a VQC, applies a trainable unitary $U(\boldsymbol{\theta})$ to the initial state $|0\rangle^{\otimes n}$ and returns a scalar expectation value of a Hermitian observable, yielding a smooth, differentiable function of $\boldsymbol{\theta}$ that acts as a trainable function approximator. Its expressibility depends on the circuit depth, qubit connectivity, and gate set³².

The PQC employed in this work (Fig. 2) operates on n qubits with L variational layers and consists of three stages:

1. *Data embedding layer.* Starting from $|0\rangle^{\otimes n}$, each qubit i receives an $R_y(\beta_i x_i)$ rotation (angle encoding³³) that encodes the i -th component of the classical input vector $\mathbf{x} \in \mathbb{R}^n$, scaled by a trainable basis parameter β_i :

$$|\psi_0\rangle = \bigotimes_{i=0}^{n-1} R_y(\beta_i x_i) |0\rangle. \quad (11)$$

2. *Variational layers.* Each of L layers $l = 1, \dots, L$ applies (i) a general single-qubit rotation $R_z(\theta_{l,i}^{(3)}) R_y(\theta_{l,i}^{(2)}) R_x(\theta_{l,i}^{(1)})$ to every qubit i , introducing $3n$ trainable parameters per layer, followed by (ii) a linear CNOT ladder $\text{CNOT}(i, i+1)$ for $i = 0, \dots, n-2$ entangling neighbouring qubits. The total number of trainable gate parameters is $3nL$, plus n basis parameters from the embedding layer.
3. *Measurement layer.* The Pauli- Z expectation values $\langle Z_i \rangle$ on each qubit are averaged to produce a single scalar output

$$f(\mathbf{x}, \boldsymbol{\beta}, \boldsymbol{\theta}) = \frac{1}{n} \sum_{i=0}^{n-1} \langle \psi | Z_i | \psi \rangle \in [-1, 1]. \quad (12)$$

The bounded output is rescaled by trainable output weights and biases in the surrounding classical layers.

4.4 JAX-based statevector simulator

We implement the PQC as a lightweight statevector simulator in JAX³⁴ rather than relying on an external framework such as PennyLane³⁵ or Qiskit³⁶. The simulator represents the quantum state as a 2^n -dimensional complex vector and applies gate operations via tensor contractions: single-qubit rotation gates (R_x, R_y, R_z) are constructed as 2×2 unitary matrices and applied with `jnp.tensordot` on the reshaped state tensor, and CNOT gates are realised through index permutation and conditional flipping on the control-target subspace. Expectation values are computed analytically from the statevector amplitudes, without stochastic sampling, and are mathematically equivalent to PennyLane’s `default.qubit` simulator: both perform exact statevector evolution under the same unitary transformations and Born-rule probabilities.

On near-term quantum hardware, gradients of PQC outputs with respect to gate parameters are typically computed via the parameter-shift rule^{*37,38}, which evaluates the circuit at two shifted parameter values per gate and therefore requires $2p$ circuit evaluations per gradient for a circuit with p parameters. In classical statevector simulation, each gate application is already a standard differentiable operation, so embedding the circuit in JAX offers two concrete advantages. First, `jax.grad` and `jax.jvp` backpropagate through the full hybrid pipeline (classical PINN layers, quantum circuit, and loss function) without parameter-shift rules or cross-framework bridging. Second, the entire training loop can be compiled into a single optimized XLA program via `jax.jit`, eliminating Python-level dispatch and framework-bridging overhead. The trade-off is that statevector simulation scales as $\mathcal{O}(2^n)$ in memory and $\mathcal{O}(Ln2^n)$ in computation, limiting the approach to moderate qubit counts; for the circuit sizes used here ($n \leq 8, L \leq 6$), this cost is negligible compared with the classical network and loss evaluation. We verify numerical correctness by comparing parameter gradients $\partial f / \partial \boldsymbol{\theta}$ against PennyLane’s `default.qubit` backend^{*39} across eight circuit configurations (2-8 qubits, 1-6 layers); the maximum absolute deviation is below 2×10^{-7} , consistent with `float32` machine precision (Supplementary Fig. S1).

4.5 Hybrid quantum-classical architecture

The hybrid architecture (Figure 1) couples two neural networks with physics-informed constraints: a wavefield network $\phi(x, z, t)$, decomposed across subdomains as described in §4.2, and a velocity network $v(x, z)$, which is a single global network. In the classical baseline, both networks are fully-connected tanh networks. In the quantum hybrid, both networks are classical-to-quantum pipelines.

Each classical-to-quantum pipeline has the form

$$g(\mathbf{x}) = w_{\text{out}} \mathcal{Q}(\mathcal{N}(\mathbf{x}; \theta_v); \boldsymbol{\beta}, \boldsymbol{\theta}_q) + b_{\text{out}}, \quad (13)$$

where $\mathcal{N}$ is a classical fully-connected tanh sub-network that maps $\mathbf{x}$ to an n -dimensional latent vector matched to the number of qubits, $\mathcal{Q}$ is the PQC of §4.3 (which returns a scalar in $[-1, 1]$), and $(w_{\text{out}}, b_{\text{out}})$ are learnable output weights. For the anomaly benchmark experiments we use $n = 4$ qubits and $L = 2$ variational layers in both networks. The velocity network output is further mapped to physical velocity units via

$$v(x, z) = v_{\text{bg}} + A \tanh(g(x, z)) \cdot m(x, z), \quad (14)$$

with background velocity $v_{\text{bg}} = 3.0$ km/s, amplitude $A = 2.0$ km/s, and a smooth spatial mask $m(x, z) \in [0, 1]$ (the product of four logistic cutoffs) that confines the velocity perturbation to the interior of the physical domain. This construction hard-codes physical bounds ($|v - v_{\text{bg}}| \leq A$) and zero perturbation at the domain edges, acting as a mild regularization of the inverse problem.

The full training objective is

$$\mathcal{L}_{\text{total}} = \lambda_{\text{data}} \mathcal{L}_{\text{data}}(\phi) + \lambda_{\text{PDE}} \mathcal{L}_{\text{PDE}}(\phi, c) + \lambda_{\text{BC}} \mathcal{L}_{\text{BC}}(\phi) + \lambda_{\text{IC}} \mathcal{L}_{\text{IC}}(\phi), \quad (15)$$

where $\mathcal{L}_{\text{data}}$ is the data misfit between the predicted and observed displacement fields at the receiver locations, $\mathcal{L}_{\text{PDE}}$ enforces the source-free acoustic wave equation (3) via automatic differentiation, and $\mathcal{L}_{\text{IC}}, \mathcal{L}_{\text{BC}}$ are the snapshot-based initial-condition and free-stress boundary-condition constraints. All parameters (classical FC weights, embedding scales β , variational angles θ_q , and output scalings) are optimized jointly by Adam gradient descent through the full differentiable pipeline.

4.6 Training setup

All training uses the Adam optimizer at learning rate 10^{-4} (baseline; see Supplementary Table S1 for variants). Each training step draws a collocation batch from the subdomains of shape $(n_x, n_z, n_t) = (40, 40, 25)$ per step (40,000 PDE collocation points in total), together with $40 \times 40 = 1,600$ points per displacement snapshot for the two initial-condition constraints, $\sim 1,000$ receiver observations (20 receivers subsampled every 100 SPEC-FEM timesteps), and 5,000 randomly-sampled boundary-condition points on the top surface. The composite loss uses fixed weights $\lambda_{\text{PDE}} = 0.1$, $\lambda_{\text{IC}} = 1.0$ (applied equally to the two snapshot constraints), $\lambda_{\text{data}} = 1.0$ (receiver data), and $\lambda_{\text{BC}} = 0.1$. The classical baseline is trained for 500,000 iterations; the quantum hybrid is trained for 65,000 iterations, at which point it has already surpassed the classical baseline’s 500,000-step L1 velocity error.

Training uses JAX’s `jit` and `vmap` transformations to compile the update step into a single XLA program and batch evaluation over subdomains. The complete update step (classical-network forward/backward passes, PQC statevector evolution, Jacobian computation via chained forward-mode `jvp`, and the composite loss) executes as a single GPU kernel. All experiments were performed on a single NVIDIA GeForce RTX 5090 GPU with `float64` precision. For the quantum hybrid, the statevector footprint is negligible: the 4-qubit PQC requires only $2^4 = 16$ complex amplitudes per forward pass, so memory cost is dominated by the classical wavefield subdomain networks.

4.7 Data

Synthetic receiver data for both benchmarks were generated with SPEC-FEM2D^{*24}, a spectral-element forward solver. Configuration files and receiver layouts are distributed with the project repository (`data/rasht/specfem/` and `data/checkerboard/specfem/`).

Anomaly benchmark. The physical domain is $1.15 \text{ km} \times 0.45 \text{ km}$ after removing a ten-element absorbing boundary layer. A single acoustic source at $(0.35, 0.19) \text{ km}$ emits a Gaussian wavelet in the SPEC-FEM forward run; 20 receivers are deployed vertically along $x = 1.15 \text{ km}$ at $\sim 22.5 \text{ m}$ spacing. Recording length is 500 ms (10,000 SPEC-FEM timesteps at $\Delta t_{\text{spec}} = 5 \times 10^{-5} \text{ s}$),

subsampled every 100 timesteps for the data misfit loss. Two wavefield snapshots at $t_0 = 0.100$ s and $t_1 = 0.115$ s (SPECFEM timesteps 2000 and 2300) are used as initial-condition constraints for the FBPINN training. Spatial coordinates are scaled by $L_x = L_z = 3$ km in all training computations.

Checkerboard benchmark. Same domain geometry, source position, and receiver layout as the anomaly benchmark, but with a 3×3 checkerboard velocity pattern of alternating ± 0.4 km/s perturbations on a 3.0 km/s background, spatially confined to $[0.70, 1.05]$ km $\times$ $[0.10, 0.30]$ km.

5 Conclusions

We introduced a hybrid quantum-classical FBPINN for acoustic FWI, integrating a PQC into the velocity network within a domain-decomposed physics-informed framework. The approach is implemented as an end-to-end differentiable JAX program, avoiding parameter-shift-rule overhead and enabling efficient joint optimization of classical PINN parameters and quantum circuit angles. On the anomaly benchmark, the hybrid model converges in $\sim 8\times$ fewer training iterations than the primary classical FBPINN baseline, despite using $\sim 33\%$ fewer trainable parameters, and it outperforms all 15 alternative classical hyperparameter variants. We emphasize that these gains are in optimization iteration count on classically-simulated circuits, and do not constitute a claim of quantum computational advantage. Key limitations include reliance on classical statevector simulation, a two-dimensional acoustic setup, and a single-source configuration; extension to elastic, three-dimensional, and multi-source inversion, as well as validation on near-term quantum hardware, are the natural next steps. The framework generalizes beyond geophysics to any wave-based inverse problem amenable to physics-informed learning.

Data availability

The synthetic receiver data used in this study were generated with SPECFEM2D ^{*24} and are available in the project repository at <https://github.com/x-repos/quFWI> under `data/rasht/` and `data/checkerboard/`.

Code availability

The JAX-based quantum simulator and quFWI framework presented here are available at <https://github.com/x-repos/quFWI>. Our implementation builds upon the FBPINNs framework ¹⁵ (<https://github.com/benmoseley/FBPINNs>) with the PINN-based FWI formulation of Rasht-Behesht et al. ¹⁴ (<https://github.com/maziarash/PINN-FWI>).

Acknowledgments

We would like to thank the Reservoir Characterization Project (RCP), Colorado School of Mines, for the financial support.

Author contributions

Conflicts of interest

The authors declare no competing interests.

References

- [1] Albert Tarantola. Inversion of seismic reflection data in the acoustic approximation. *Geophysics*, 49(8):1259–1266, 08 1984. ISSN 0016-8033. doi: 10.1190/1.1441754. URL <https://doi.org/10.1190/1.1441754>.
- [2] Andreas Fichtner, Brian L. N. Kennett, Heiner Igel, and Hans-Peter Bunge. Full seismic waveform tomography for upper-mantle structure in the australasian region using adjoint methods. *Geophysical Journal International*, 179(3):1703–1725, 12 2009. ISSN 0956-540X. doi: 10.1111/j.1365-246X.2009.04368.x. URL <https://doi.org/10.1111/j.1365-246X.2009.04368.x>.
- [3] Ebru Bozdağ, Jeannot Trampert, and Jeroen Tromp. Misfit functions for full waveform inversion based on instantaneous phase and envelope measurements. *Geophysical Journal International*, 185(2):845–870, 05 2011. ISSN 0956-540X. doi: 10.1111/j.1365-246X.2011.04970.x. URL <https://doi.org/10.1111/j.1365-246X.2011.04970.x>.
- [4] J. Virieux and S. Operto. An overview of full-waveform inversion in exploration geophysics. *Geophysics*, 74(6):WCC1–WCC26, 12 2009. ISSN 0016-8033. doi: 10.1190/1.3238367. URL <https://doi.org/10.1190/1.3238367>.
- [5] Zhongwen Zhan. Distributed acoustic sensing turns fiber-optic cables into sensitive seismic antennas. *Seismological Research Letters*, 91(1):1–15, 2020.
- [6] J. Wiskin, D. T. Borup, S. A. Johnson, and M. Berggren. Non-linear inverse scattering: High resolution quantitative breast tissue tomography. *The Journal of the Acoustical Society of America*, 131(5):3802–3813, 05 2012. ISSN 0001-4966. doi: 10.1121/1.3699240. URL <https://doi.org/10.1121/1.3699240>.
- [7] G Y Sandhu, C Li, O Roy, S Schmidt, and N Duric. Frequency domain ultrasound waveform tomography: breast imaging using a ring transducer. *Physics in Medicine & Biology*, 60(14):5381, jun 2015. doi: 10.1088/0031-9155/60/14/5381. URL <https://doi.org/10.1088/0031-9155/60/14/5381>.
- [8] Lihong V. Wang and Junjie Yao. A practical guide to photoacoustic tomography in the life sciences. *Nature Methods*, 13(8):627–638, 08 2016. doi: 10.1038/nmeth.3925. URL <https://doi.org/10.1038/nmeth.3925>.
- [9] Jing Rao, Madis Ratasseppe, and Zheng Fan. Guided wave tomography based on full waveform inversion. *IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control*, 63(5):737–745, 2016. doi: 10.1109/TUFFC.2016.2536144.
- [10] M. Raissi, P. Perdikaris, and G.E. Karniadakis. Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational Physics*, 378:686–707, 2019. ISSN 0021-9991. doi: <https://doi.org/10.1016/j.jcp.2018.10.045>. URL <https://www.sciencedirect.com/science/article/pii/S0021999118307125>.
- [11] Chao Song, Tariq Alkhalifah, and Umair Bin Waheed. Solving the frequency-domain acoustic vti wave equation using physics-informed neural networks. *Geophysical Journal International*, 225(2):846–859, 01 2021. ISSN 0956-540X. doi: 10.1093/gji/ggab010. URL <https://doi.org/10.1093/gji/ggab010>.
- [12] Umair bin Waheed, Tariq Alkhalifah, Ehsan Haghighat, Chao Song, and Jean Virieux. Pinntomo: Seismic tomography using physics-informed neural networks, 2021. URL <https://arxiv.org/abs/2104.01588>.

- [13] Xilun Liu and Mohamed Almekkawy. Full wave inversion for ultrasound tomography using physics based deep neural network. In *2023 IEEE 20th International Symposium on Biomedical Imaging (ISBI)*, pages 1–4, 2023. doi: 10.1109/ISBI53787.2023.10230537.
- [14] Majid Rasht-Behesht, Christian Huber, Khemraj Shukla, and George Em Karniadakis. Physics-informed neural networks (pinns) for wave propagation and full waveform inversions. *Journal of Geophysical Research: Solid Earth*, 127(5), 2022. doi: <https://doi.org/10.1029/2021JB023120>. URL <https://agupubs.onlinelibrary.wiley.com/doi/abs/10.1029/2021JB023120>.
- [15] Ben Moseley, Andrew Markham, and Tarje Nissen-Meyer. Finite basis physics-informed neural networks (FBPINNs): a scalable domain decomposition approach for solving differential equations. *Advances in Computational Mathematics*, 49(4):62, 07 2023. doi: 10.1007/s10444-023-10065-9. URL <https://doi.org/10.1007/s10444-023-10065-9>.
- [16] John Preskill. Quantum Computing in the NISQ era and beyond. *Quantum*, 2:79, August 2018. ISSN 2521-327X. doi: 10.22331/q-2018-08-06-79. URL <https://doi.org/10.22331/q-2018-08-06-79>.
- [17] Maria Schuld, Alex Bocharov, Krysta M. Svore, and Nathan Wiebe. Circuit-centric quantum classifiers. *Phys. Rev. A*, 101:032308, Mar 2020. doi: 10.1103/PhysRevA.101.032308. URL <https://link.aps.org/doi/10.1103/PhysRevA.101.032308>.
- [18] Marcello Benedetti, Erika Lloyd, Stefan Sack, and Mattia Fiorentini. Parameterized quantum circuits as machine learning models. *Quantum Science and Technology*, 4(4):043001, nov 2019. doi: 10.1088/2058-9565/ab4eb5. URL <https://doi.org/10.1088/2058-9565/ab4eb5>.
- [19] Mateusz Ostaszewski, Edward Grant, and Marcello Benedetti. Structure optimization for parameterized quantum circuits. *Quantum*, 5:391, January 2021. ISSN 2521-327X. doi: 10.22331/q-2021-01-28-391. URL <https://doi.org/10.22331/q-2021-01-28-391>.
- [20] Annie E. Paine, Vincent E. Elfving, and Oleksandr Kyriienko. Quantum quantile mechanics: Solving stochastic differential equations for generating time-series. *Advanced Quantum Technologies*, 6(10):2300065, 2023. doi: <https://doi.org/10.1002/qute.202300065>. URL <https://advanced.onlinelibrary.wiley.com/doi/abs/10.1002/qute.202300065>.
- [21] Fong Yew Leong, Wei-Bin Ewe, Si Bui Quang Tran, Zhongyuan Zhang, and Jun Yong Khoo. Hybrid quantum physics-informed neural network: Towards efficient learning of high-speed flows. *Computers & Fluids*, 301:106782, 2025. ISSN 0045-7930. doi: <https://doi.org/10.1016/j.compfluid.2025.106782>. URL <https://www.sciencedirect.com/science/article/pii/S0045793025002427>.
- [22] Stefan Berger, Norbert Hosters, and Matthias Möller. Trainable embedding quantum physics informed neural networks for solving nonlinear PDEs. *Scientific Reports*, 15(1):18823, 05 2025. doi: 10.1038/s41598-025-02959-z. URL <https://doi.org/10.1038/s41598-025-02959-z>.
- [23] Hoang Anh Nguyen and Ali Tura. Seismic traveltime inversion with quantum annealing. *Scientific Reports*, 15(1):17984, 2025. doi: 10.1038/s41598-025-01188-8.
- [24] Dimitri Komatitsch and Jean-Pierre Vilotte. The spectral element method: An efficient tool to simulate the seismic response of 2d and 3d geological structures. *Bulletin of the Seismological Society of America*, 88(2):368–392, 1998.
- [25] M. Cerezo, Andrew Arrasmith, Ryan Babbush, Simon C. Benjamin, Suguru Endo, Keisuke Fujii, Jarrod R. McClean, Kosuke Mitarai, Xiao Yuan, Lukasz Cincio, and Patrick J. Coles. Variational quantum algorithms. *Nature Reviews Physics*, 3(9):625–644, 2021. doi: 10.1038/s42254-021-00348-9.

- [26] Vojtěch Havlíček, Antonio D. Córcoles, Kristan Temme, Aram W. Harrow, Abhinav Kandala, Jerry M. Chow, and Jay M. Gambetta. Supervised learning with quantum-enhanced feature spaces. *Nature*, 567(7747):209–212, 2019. doi: 10.1038/s41586-019-0980-2.
- [27] Hsin-Yuan Huang, Michael Broughton, Masoud Mohseni, Ryan Babbush, Sergio Boixo, Hartmut Neven, and Jarrod R. McClean. Power of data in quantum machine learning. *Nature Communications*, 12(1):2631, 2021. doi: 10.1038/s41467-021-22539-9.
- [28] Maria Schuld, Ryan Sweke, and Johannes Jakob Meyer. Effect of data encoding on the expressive power of variational quantum-machine-learning models. *Physical Review A*, 103(3):032430, 2021. doi: 10.1103/PhysRevA.103.032430.
- [29] Jarrod R. McClean, Sergio Boixo, Vadim N. Smelyanskiy, Ryan Babbush, and Hartmut Neven. Barren plateaus in quantum neural network training landscapes. *Nature Communications*, 9(1):4812, 2018. doi: 10.1038/s41467-018-07090-4.
- [30] Aditi Krishnapriyan, Amir Gholami, Shandian Zhe, Robert Kirby, and Michael W. Mahoney. Characterizing possible failure modes in physics-informed neural networks. In *Advances in Neural Information Processing Systems*, volume 34, pages 26548–26560, 2021.
- [31] Marcello Benedetti, Erika Lloyd, Stefan Sack, and Mattia Fiorentini. Parameterized quantum circuits as machine learning models. *Quantum Science and Technology*, 4(4):043001, 2019. doi: 10.1088/2058-9565/ab4eb5.
- [32] Sukin Sim, Peter D. Johnson, and Alán Aspuru-Guzik. Expressibility and entangling capability of parameterized quantum circuits for hybrid quantum-classical algorithms. *Advanced Quantum Technologies*, 2(12):1900070, 2019. doi: 10.1002/qute.201900070.
- [33] Maria Schuld and Francesco Petruccione. *Machine Learning with Quantum Computers*. Quantum Science and Technology. Springer, 2nd edition, 2021.
- [34] James Bradbury, Roy Frostig, Peter Hawkins, Matthew James Johnson, Chris Leary, Dougal Maclaurin, George Necula, Adam Paszke, Jake VanderPlas, Skye Wanderman-Milne, and Qiao Zhang. JAX: composable transformations of Python+NumPy programs, 2018. URL <http://github.com/jax-ml/jax>.
- [35] Ville Bergholm, Josh Izaac, Maria Schuld, Christian Gogolin, Shahnawaz Ahmed, Vishnu Ajith, M. Sohaib Alam, Guillermo Alonso-Linaje, B. AkashNarayanan, Ali Asadi, Juan Miguel Arrazola, Utkarsh Azad, Sam Banning, Carsten Blank, Thomas R Bromley, Benjamin A. Cordier, Jack Ceroni, Alain Delgado, Olivia Di Matteo, Amintor Dusko, Tanya Garg, Diego Guala, Anthony Hayes, Ryan Hill, Aroosa Ijaz, Theodor Isacsson, David Ittah, Soran Jahangiri, Prateek Jain, Edward Jiang, Ankit Khandelwal, Korbinian Kottmann, Robert A. Lang, Christina Lee, Thomas Loke, Angus Lowe, Keri McKiernan, Johannes Jakob Meyer, J. A. Montañez-Barrera, Romain Moyard, Zeyue Niu, Lee James O’Riordan, Steven Oud, Ashish Panigrahi, Chae-Yeun Park, Daniel Polatajko, Nicolás Quesada, Chase Roberts, Nahum Sá, Isidor Schoch, Borun Shi, Shuli Shu, Sukin Sim, Arshpreet Singh, Ingrid Strandberg, Jay Soni, Antal Száva, Slimane Thabet, Rodrigo A. Vargas-Hernández, Trevor Vincent, Nicola Vitucci, Maurice Weber, David Wierichs, Roeland Wiersema, Moritz Willmann, Vincent Wong, Shaoming Zhang, and Nathan Killoran. PennyLane: Automatic differentiation of hybrid quantum-classical computations, 2022. URL <https://arxiv.org/abs/1811.04968>.
- [36] Ali Javadi-Abhari, Matthew Treinish, Kevin Krsulich, Christopher J. Wood, Jake Lishman, Julien Gacon, Simon Martiel, Paul D. Nation, Lev S. Bishop, Andrew W. Cross, Blake R. Johnson, and Jay M. Gambetta. Quantum computing with qiskit, 2024. URL <https://arxiv.org/abs/2405.08810>.

- [37] Kosuke Mitarai, Makoto Negoro, Masahiro Kitagawa, and Keisuke Fujii. Quantum circuit learning. *Physical Review A*, 98(3):032309, 2018. doi: 10.1103/PhysRevA.98.032309.
- [38] Maria Schuld, Ville Bergholm, Christian Gogolin, Josh Izaac, and Nathan Killoran. Evaluating analytic gradients on quantum hardware. *Physical Review A*, 99(3):032331, 2019. doi: 10.1103/PhysRevA.99.032331.
- [39] Ville Bergholm, Josh Izaac, Maria Schuld, Christian Gogolin, Shahnawaz Ahmed, Vishnu Ajith, M. Sohaib Alam, Guillermo Alonso-Linaje, B. AkashNarayanan, Ali Asadi, et al. PennyLane: Automatic differentiation of hybrid quantum-classical computations, 2018.

550